

Statistical Computing

Probability and Statistics Review

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Outline

Random Variables and Probability

Some Discrete Distributions

Some Continuous Distributions

Multivariate Normal Distribution

Statistics

Random Variables and Probability

Cumulative Distribution Function (CDF)

Definition

For a random variable X , the cumulative distribution function (cdf) is

$$F_X(x) = \mathbb{P}(X \leq x), \quad x \in \mathbb{R}. \quad (1)$$

When the context is clear, we write $F(x)$.

Cumulative Distribution Function (CDF)

Basic properties

1. F_X is non-decreasing.
2. F_X is right-continuous:

$$\lim_{\varepsilon \rightarrow 0^+} F_X(x + \varepsilon) = F_X(x), \quad \forall x \in \mathbb{R}. \quad (2)$$

3. Boundary limits:

$$\lim_{x \rightarrow -\infty} F_X(x) = 0, \quad \lim_{x \rightarrow \infty} F_X(x) = 1. \quad (3)$$

Discrete Random Variables: PMF and CDF

PMF

If X is discrete, its probability mass function (pmf) is

$$p_X(x) = \mathbb{P}(X = x). \quad (4)$$

At a discontinuity point,

$$p_X(x) = F_X(x) - F_X(x^-), \quad F_X(x^-) := \lim_{t \rightarrow x^-} F_X(t). \quad (5)$$

CDF as a cumulative sum

$$F_X(x) = \mathbb{P}(X \leq x) = \sum_{\{k \leq x: p_X(k) > 0\}} p_X(k). \quad (6)$$

PDF

If F_X is differentiable, the probability density function (pdf) is

$$f_X(x) = \frac{d}{dx} F_X(x). \quad (7)$$

CDF as an integral of the density

$$F_X(x) = \mathbb{P}(X \leq x) = \int_{-\infty}^x f_X(t) dt. \quad (8)$$

Joint and Marginal Distributions

Joint distribution (continuous case)

For (X, Y) with joint pdf $f_{X,Y}$,

$$F_{X,Y}(x, y) = \mathbb{P}(X \leq x, Y \leq y) = \int_{-\infty}^y \int_{-\infty}^x f_{X,Y}(s, t) ds dt. \quad (9)$$

Marginal densities

$$f_X(x) = \int_{-\infty}^{\infty} f_{X,Y}(x, y) dy, \quad f_Y(y) = \int_{-\infty}^{\infty} f_{X,Y}(x, y) dx. \quad (10)$$

(The discrete case replaces integrals with sums.)

Support of a density

$$\text{supp}(X) = \{x : f_X(x) > 0\}, \quad \text{supp}(X, Y) = \{(x, y) : f_{X,Y}(x, y) > 0\}. \quad (11)$$

Expectation and Moments

Expectation

For continuous X ,

$$\mathbb{E}[X] = \int_{-\infty}^{\infty} x f_X(x) dx. \quad (12)$$

For discrete X ,

$$\mathbb{E}[X] = \sum_{\{x: p_X(x) > 0\}} x p_X(x). \quad (13)$$

Expectation of a function; moments

For suitable g ,

$$\mathbb{E}[g(X)] = \int_{-\infty}^{\infty} g(x) f_X(x) dx \quad (\text{continuous}). \quad (14)$$

The r -th moment is $\mathbb{E}[X^r]$.

Variance and Standard Deviation

Variance

Let $\mu_X = \mathbb{E}[X]$. The variance is the 2nd central moment:

$$\text{Var}(X) = \mathbb{E}[(X - \mathbb{E}[X])^2]. \quad (15)$$

Equivalently,

$$\text{Var}(X) = \mathbb{E}[X^2] - (\mathbb{E}[X])^2 = \mathbb{E}[X^2] - \mu_X^2. \quad (16)$$

Related quantities

Standard deviation: $\sqrt{\text{Var}(X)}$. Precision: $\text{Var}(X)^{-1}$.

Covariance

$$\text{Cov}(X, Y) = \mathbb{E}[(X - \mu_X)(Y - \mu_Y)] = \mathbb{E}[XY] - \mathbb{E}[X]\mathbb{E}[Y]. \quad (17)$$

Correlation

$$\text{Corr}(X, Y) = \frac{\text{Cov}(X, Y)}{\sqrt{\text{Var}(X) \text{Var}(Y)}} = \mathbb{E} \left[\frac{X - \mu_X}{\sqrt{\text{Var}(X)}} \cdot \frac{Y - \mu_Y}{\sqrt{\text{Var}(Y)}} \right]. \quad (18)$$

Conditional Probability (Events)

Definition

For events A, B with $\mathbb{P}(B) > 0$,

$$\mathbb{P}(A \mid B) = \frac{\mathbb{P}(A \cap B)}{\mathbb{P}(B)}. \quad (19)$$

Product rule

$$\mathbb{P}(A \cap B) = \mathbb{P}(A \mid B)\mathbb{P}(B) = \mathbb{P}(B \mid A)\mathbb{P}(A). \quad (20)$$

Conditional density (continuous)

$$f_{X|Y=y}(x) = \frac{f_{X,Y}(x,y)}{f_Y(y)}, \quad f_{Y|X=x}(y) = \frac{f_{X,Y}(x,y)}{f_X(x)}. \quad (21)$$

Factorization

$$f_{X,Y}(x,y) = f_{X|Y=y}(x) f_Y(y) = f_{Y|X=x}(y) f_X(x). \quad (22)$$

Independence and i.i.d. Sampling

Independence of two random variables

X and Y are independent iff

$$f_{X,Y}(x,y) = f_X(x)f_Y(y) \quad (\forall x,y), \quad (23)$$

equivalently $F_{X,Y}(x,y) = F_X(x)F_Y(y)$ for all (x,y) .

Independence in d dimensions

For $X_1, \dots, X_d$ with joint density (or pmf) f ,

$$X_1, \dots, X_d \text{ independent} \iff f(x_1, \dots, x_d) = \prod_{j=1}^d f_j(x_j). \quad (24)$$

i.i.d. sample

If $X_1, \dots, X_n$ are i.i.d. from f_X ,

$$f(x_1, \dots, x_n) = \prod_{i=1}^n f_X(x_i). \quad (25)$$

Useful Identities: Linearity and Affine Transforms

Linearity of expectation

For constants a, b ,

$$\mathbb{E}[aX + b] = a\mathbb{E}[X] + b, \quad \mathbb{E}[X + Y] = \mathbb{E}[X] + \mathbb{E}[Y]. \quad (26)$$

If $X \perp Y$, then $\mathbb{E}[XY] = \mathbb{E}[X]\mathbb{E}[Y]$.

Variance under affine transforms

$$\text{Var}(b) = 0, \quad \text{Var}(aX + b) = a^2 \text{Var}(X). \quad (27)$$

Variance of a sum

$$\text{Var}(X + Y) = \text{Var}(X) + \text{Var}(Y) + 2 \text{Cov}(X, Y). \quad (28)$$

If $X \perp Y$, then $\text{Var}(X + Y) = \text{Var}(X) + \text{Var}(Y)$.

Useful Identities: Sums and Sample Mean (cont.)

Sum and sample mean

If $X_1, \dots, X_n$ are i.i.d. with mean μ and variance σ^2 ,

$$\mathbb{E}\left[\sum_{i=1}^n X_i\right] = n\mu, \quad \text{Var}\left(\sum_{i=1}^n X_i\right) = n\sigma^2. \quad (29)$$

For $\bar{X} = \frac{1}{n} \sum_{i=1}^n X_i$,

$$\mathbb{E}[\bar{X}] = \mu, \quad \text{Var}(\bar{X}) = \frac{\sigma^2}{n}. \quad (30)$$

Some Discrete Distributions

Bernoulli(p)

A single success/failure trial:

$$\mathbb{P}(X = 1) = p, \quad \mathbb{P}(X = 0) = 1 - p. \quad (31)$$

$$\mathbb{E}[X] = p, \quad \text{Var}(X) = p(1 - p). \quad (32)$$

Bernoulli trials

If $X_1, X_2, \dots$ are independent and identically distributed Bernoulli(p), we call them Bernoulli trials.

Binomial Distribution

Binomial(n, p)

Let X be the number of successes in n i.i.d. Bernoulli(p) trials:

$$\mathbb{P}(X = x) = \binom{n}{x} p^x (1 - p)^{n-x}, \quad x = 0, 1, \dots, n. \quad (33)$$

Mean and variance

Viewing $X = \sum_{i=1}^n X_i$ with $X_i \sim \text{Bernoulli}(p)$,

$$\mathbb{E}[X] = np, \quad \text{Var}(X) = np(1 - p). \quad (34)$$

Geometric Distribution

Geometric(p): number of failures before the first success

In Bernoulli(p) trials, let X be the number of failures before the first success:

$$\mathbb{P}(X = x) = p(1 - p)^x, \quad x = 0, 1, 2, \dots \quad (35)$$

CDF, mean, variance

For integer $x \geq 0$,

$$\mathbb{P}(X \leq x) = 1 - (1 - p)^{x+1}. \quad (36)$$

$$\mathbb{E}[X] = \frac{1 - p}{p}, \quad \text{Var}(X) = \frac{1 - p}{p^2}. \quad (37)$$

Some Continuous Distributions

Normal distribution

Definition

A normal random variable with mean μ and variance σ^2 (denoted $N(\mu, \sigma^2)$) has pdf

$$f(x) = \frac{1}{\sqrt{2\pi}\sigma} \exp\left(-\frac{1}{2} \left(\frac{x - \mu}{\sigma}\right)^2\right), \quad -\infty < x < \infty. \quad (38)$$

Standard normal CDF

The standard normal is $N(0, 1)$. Its cdf is

$$\Phi(z) = \int_{-\infty}^z \frac{1}{\sqrt{2\pi}} e^{-t^2/2} dt, \quad -\infty < z < \infty. \quad (39)$$

Normal distribution: linear transforms and standardization

Linear transformation

If $X \sim N(\mu, \sigma^2)$ and $Y = aX + b$, then

$$Y \sim N(a\mu + b, a^2\sigma^2). \quad (40)$$

Standardization

If $X \sim N(\mu, \sigma^2)$, then

$$Z = \frac{X - \mu}{\sigma} \sim N(0, 1). \quad (41)$$

Normal distribution: linear combinations

Linear combinations

If $X_1, \dots, X_k$ are independent with $X_i \sim N(\mu_i, \sigma_i^2)$ and $a_1, \dots, a_k$ are constants, then $Y = \sum_{i=1}^k a_i X_i$ is normal with

$$\mathbb{E}[Y] = \sum_{i=1}^k a_i \mu_i, \quad \text{Var}(Y) = \sum_{i=1}^k a_i^2 \sigma_i^2. \quad (42)$$

Sampling from a normal population

If $X_1, \dots, X_n \stackrel{\text{iid}}{\sim} N(\mu, \sigma^2)$, then

$$\sum_{i=1}^n X_i \sim N(n\mu, n\sigma^2), \quad \bar{X} = \frac{1}{n} \sum_{i=1}^n X_i \sim N\left(\mu, \frac{\sigma^2}{n}\right). \quad (43)$$

Gamma distribution

Gamma(r, λ)

A random variable X has a gamma distribution with shape $r > 0$ and rate $\lambda > 0$ if its pdf is

$$f(x) = \frac{\lambda^r}{\Gamma(r)} x^{r-1} e^{-\lambda x}, \quad x \geq 0, \quad (44)$$

where $\Gamma(\cdot)$ is the complete gamma function.

Gamma function

$$\Gamma(r) = \int_0^\infty t^{r-1} e^{-t} dt, \quad r \neq 0, -1, -2, \dots \quad (45)$$

For positive integers n , $\Gamma(n) = (n-1)!$.

Moments

If $X \sim \text{Gamma}(r, \lambda)$, then

$$\mathbb{E}[X] = \frac{r}{\lambda}, \quad \text{Var}(X) = \frac{r}{\lambda^2}. \quad (46)$$

Exponential distribution

Exponential(λ)

The exponential distribution is the special case of gamma with $r = 1$. If $X \sim \text{Exp}(\lambda)$ (rate $\lambda > 0$), then

$$f(x) = \lambda e^{-\lambda x}, \quad x \geq 0, \quad (47)$$

$$\mathbb{E}[X] = \frac{1}{\lambda}, \quad \text{Var}(X) = \frac{1}{\lambda^2}. \quad (48)$$

Sum of exponentials

If $X_1, \dots, X_r$ are i.i.d. $\text{Exp}(\lambda)$, then

$$\sum_{i=1}^r X_i \sim \text{Gamma}(r, \lambda). \quad (49)$$

Chi-square distribution

Definition

The chi-square distribution with ν degrees of freedom (denoted $\chi^2(\nu)$) has pdf

$$f(x) = \frac{1}{\Gamma(\nu/2) 2^{\nu/2}} x^{\nu/2-1} e^{-x/2}, \quad x \geq 0, \nu = 1, 2, \dots \quad (50)$$

Gamma connection

$\chi^2(\nu)$ is a special case of $\text{Gamma}(r, \lambda)$ with $r = \nu/2$ and $\lambda = 1/2$.

Chi-square: basic properties

Normal-square characterization

If $Z \sim N(0, 1)$, then $Z^2 \sim \chi^2(1)$. If $Z_1, \dots, Z_\nu$ are i.i.d. $N(0, 1)$, then

$$\sum_{i=1}^{\nu} Z_i^2 \sim \chi^2(\nu). \quad (51)$$

Additivity and moments

If $X \sim \chi^2(\nu_1)$ and $Y \sim \chi^2(\nu_2)$ are independent, then

$$X + Y \sim \chi^2(\nu_1 + \nu_2). \quad (52)$$

Also, if $X \sim \chi^2(\nu)$,

$$\mathbb{E}[X] = \nu, \quad \text{Var}(X) = 2\nu. \quad (53)$$

Beta and Uniform distributions

Beta(α, β)

A random variable X has a Beta distribution with $\alpha > 0, \beta > 0$ if

$$f(x) = \frac{\Gamma(\alpha + \beta)}{\Gamma(\alpha)\Gamma(\beta)} x^{\alpha-1} (1-x)^{\beta-1}, \quad 0 \leq x \leq 1. \quad (54)$$

Beta function

The normalizing constant is the reciprocal of the beta function

$$B(\alpha, \beta) = \int_0^1 t^{\alpha-1} (1-t)^{\beta-1} dt = \frac{\Gamma(\alpha)\Gamma(\beta)}{\Gamma(\alpha + \beta)}. \quad (55)$$

Beta: mean/variance and the Uniform special case

Mean and variance

If $X \sim \text{Beta}(\alpha, \beta)$, then

$$\mathbb{E}[X] = \frac{\alpha}{\alpha + \beta}, \quad \text{Var}(X) = \frac{\alpha\beta}{(\alpha + \beta)^2(\alpha + \beta + 1)}. \quad (56)$$

Uniform(0,1)

Uniform(0, 1) is the special case Beta(1, 1), hence

$$\mathbb{E}[X] = \frac{1}{2}, \quad \text{Var}(X) = \frac{1}{12}. \quad (57)$$

Multivariate Normal Distribution

Definition

Two continuous random variables (X, Y) have a bivariate normal distribution if their joint pdf is

$$f(x, y) = \frac{1}{2\pi\sigma_1\sigma_2\sqrt{1-\rho^2}} \exp\left(-\frac{1}{2(1-\rho^2)} \left[\left(\frac{x-\mu_1}{\sigma_1}\right)^2 - 2\rho\left(\frac{x-\mu_1}{\sigma_1}\right)\left(\frac{y-\mu_2}{\sigma_2}\right) + \left(\frac{y-\mu_2}{\sigma_2}\right)^2 \right]\right), \quad (58)$$

for $(x, y) \in \mathbb{R}^2$.

Parameters

$\mu_1 = \mathbb{E}[X]$, $\mu_2 = \mathbb{E}[Y]$, $\sigma_1^2 = \text{Var}(X)$, $\sigma_2^2 = \text{Var}(Y)$, and $\rho = \text{Corr}(X, Y)$.

Bivariate normal: key properties

Marginals

If (X, Y) is bivariate normal, then

$$X \sim N(\mu_1, \sigma_1^2), \quad Y \sim N(\mu_2, \sigma_2^2). \quad (59)$$

Conditionals

The conditional distribution of Y given $X = x$ is normal with

$$Y \mid X = x \sim N\left(\mu_2 + \rho \frac{\sigma_2}{\sigma_1}(x - \mu_1), \sigma_2^2(1 - \rho^2)\right). \quad (60)$$

Similarly,

$$X \mid Y = y \sim N\left(\mu_1 + \rho \frac{\sigma_1}{\sigma_2}(y - \mu_2), \sigma_1^2(1 - \rho^2)\right). \quad (61)$$

Independence

For the bivariate normal distribution,

$$X \text{ and } Y \text{ are independent} \iff \rho = 0. \quad (62)$$

Definition

A random vector $X = (X_1, \dots, X_d)^T$ is multivariate normal, denoted $N_d(\mu, \Sigma)$, if its joint pdf is

$$f(x) = \frac{1}{(2\pi)^{d/2} |\Sigma|^{1/2}} \exp\left(-\frac{1}{2}(x - \mu)^T \Sigma^{-1}(x - \mu)\right), \quad x \in \mathbb{R}^d, \quad (63)$$

where $\mu \in \mathbb{R}^d$ is the mean vector and Σ is a nonsingular covariance matrix.

Multivariate normal: marginals and independence

Marginals

Each one-dimensional marginal X_i is normal with mean μ_i and variance Σ_{ii} . More generally, all marginal distributions of a multivariate normal vector are multivariate normal.

Independence

The components $X_1, \dots, X_d$ are independent if and only if Σ is diagonal.

Affine mapping

Let C be an $m \times d$ matrix and $b \in \mathbb{R}^m$. If $X \sim N_d(\mu, \Sigma)$ and

$$Y = CX + b, \tag{64}$$

then Y is multivariate normal with

$$Y \sim N_m(C\mu + b, C\Sigma C^T). \tag{65}$$

Statistics

Notation

Unless stated otherwise, $X_1, \dots, X_n$ denotes a random sample from a distribution with cdf $F(x)$, pdf/pmf $f(x)$, mean μ , and variance σ^2 . Lowercase $x_1, \dots, x_n$ denote an observed sample.

Statistic

A statistic is any function of the sample:

$$T_n = T(X_1, \dots, X_n). \quad (66)$$

Sample mean and sample variance

Sample mean

$$\bar{X} = \frac{1}{n} \sum_{i=1}^n X_i. \quad (67)$$

Sample variance

$$S^2 = \frac{1}{n-1} \sum_{i=1}^n (X_i - \bar{X})^2 = \frac{\sum_{i=1}^n X_i^2 - n\bar{X}^2}{n-1}. \quad (68)$$

The sample standard deviation is $S = \sqrt{S^2}$.

Definition

An estimator of $F(x) = \mathbb{P}(X \leq x)$ is the proportion of sample points that fall in $(-\infty, x]$. The empirical cdf (ecdf) is

$$F_n(x) = \frac{1}{n} \sum_{i=1}^n \mathbf{1}\{X_i \leq x\}. \quad (69)$$

Piecewise form

Let $x_{(1)} \leq x_{(2)} \leq \cdots \leq x_{(n)}$ be the ordered sample. Then

$$F_n(x) = \begin{cases} 0, & x < x_{(1)}, \\ \frac{i}{n}, & x_{(i)} \leq x < x_{(i+1)}, \quad i = 1, \dots, n-1, \\ 1, & x_{(n)} \leq x. \end{cases} \quad (70)$$

Population quantile

A quantile is defined by inverting the cdf. Since F may not be strictly increasing, the q -quantile ($0 < q < 1$) is

$$x_q = \inf\{x \in \mathbb{R} : F(x) \geq q\}. \quad (71)$$

Sample quantiles

Quantiles can be estimated using the inverse of the ecdf or other functions of order statistics.

Unbiased and asymptotically unbiased

An estimator $\hat{\theta}_n$ is unbiased for θ if

$$\mathbb{E}[\hat{\theta}_n] = \theta. \quad (72)$$

It is asymptotically unbiased if

$$\lim_{n \rightarrow \infty} \mathbb{E}[\hat{\theta}_n] = \theta. \quad (73)$$

Bias

$$\text{bias}(\hat{\theta}) = \mathbb{E}[\hat{\theta}] - \theta. \quad (74)$$

Mean squared error

$$\text{MSE}(\hat{\theta}) = \mathbb{E}[(\hat{\theta} - \theta)^2]. \quad (75)$$

Variance–bias decomposition

$$\text{MSE}(\hat{\theta}) = \text{Var}(\hat{\theta}) + (\text{bias}(\hat{\theta}))^2. \quad (76)$$

In particular, if $\hat{\theta}$ is unbiased, then $\text{MSE}(\hat{\theta}) = \text{Var}(\hat{\theta})$.

Standard error: key examples

Standard error

The standard error of an estimator $\hat{\theta}$ is

$$\text{se}(\hat{\theta}) = \sqrt{\text{Var}(\hat{\theta})}. \quad (77)$$

Examples

- Mean: if $\text{Var}(X) = \sigma^2$, then $\text{se}(\bar{X}) = \sqrt{\text{Var}(\bar{X})} = \frac{\sigma}{\sqrt{n}}$.
- Proportion: $\text{se}(\hat{p}) = \sqrt{p(1-p)/n} \leq 0.5/\sqrt{n}$.
- For fixed x , $F_n(x)$ is unbiased for $F(x)$ and

$$\text{se}(F_n(x)) = \sqrt{F(x)(1-F(x))/n} \leq 0.5/\sqrt{n}. \quad (78)$$

What we now have

- Probability review: distributions, conditioning, and key families (discrete/continuous, multivariate normal).
- Estimation basics: statistics, bias/MSE, and standard error.